AGENTIC-ARBITER

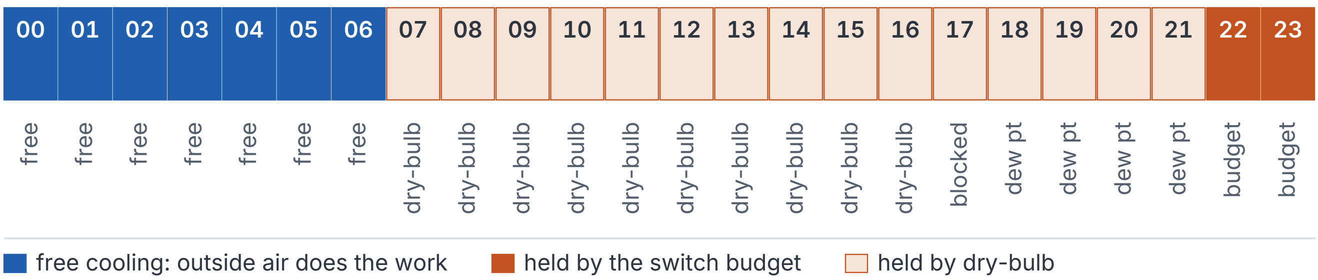
Free-cooling decision report · Amazon Web Services, OR · station KRLD · 2024-06-02

An agent decides, hour by hour, whether this data centre can switch its mechanical chillers off and cool with outside air instead. Every hour it releases carries a safety margin measured from the agent's own past errors, and it declines the hours it cannot stand behind.

$306k – $1254k at this site: $305,846 to $1,254,023, at $5,543 to $11,365 per MW of IT load per year MODELED	**14.3%** less mechanical cooling runtime, a share so it holds at any hall size	**+407 h** chiller-hours recovered per year against the control operators run today	**42,730** hours of real recorded weather, 896 days held out	**65.6%** bound coverage on the four measured forecast pairs, against a 90% target

The agent's day, hour by hour

7 of 24 hours free cooling



What this day shows

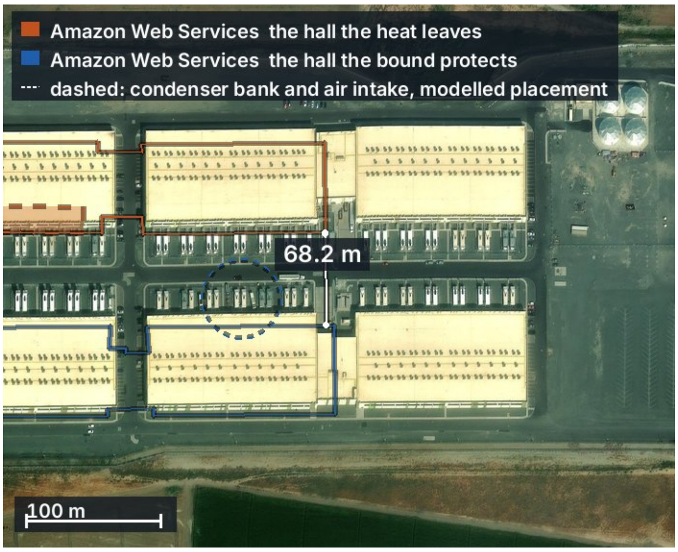
- Just 7 of the 24 hours ran on outside air. The remaining 17 ran chillers, and every one of them has a named reason rather than a default.
- **One constraint** shaped the day: Blocked by air temperature accounts for 10 of the 24 hours, 07:00 to 16:00.
- **The bound held** across 896 days the agent never trained on. It took 16,559 free-cooling hours and 15 of them turned out unsafe: 0.91 per thousand.
- **The forecast is the product**: hold everything else and drop the forecast to no skill at all, and the 407 hours a year this recovers becomes 40.

What makes this different from a thermostat with a forecast attached: where the site geometry defeats the physics, the agent refuses the hour rather than return a number it cannot stand behind, and the cost of that caution is measured and published rather than hidden.

Contents

This document is one site's complete analysis. Page 1 is the whole argument; everything after it is the evidence.

The site



Two halls, Amazon Web Services / Amazon Web Services. The agent is deciding for the second one. When its neighbour rejects heat from the condensers along the facing wall, that warm air has only **68.2 m** of open ground to cross before it reaches the air this building breathes.

That is what a **plume** is: a moving body of warmer air leaving one building and drifting with the wind. It matters because free cooling only works while the air arriving at the intake is genuinely cool. A thermostat reading the regional forecast cannot see this, because the forecast is for the region and the plume is a hundred metres wide.

So the agent solves for it on this footprint, at this separation, for every wind bearing, and carries the answer as extra margin before it will release an hour.

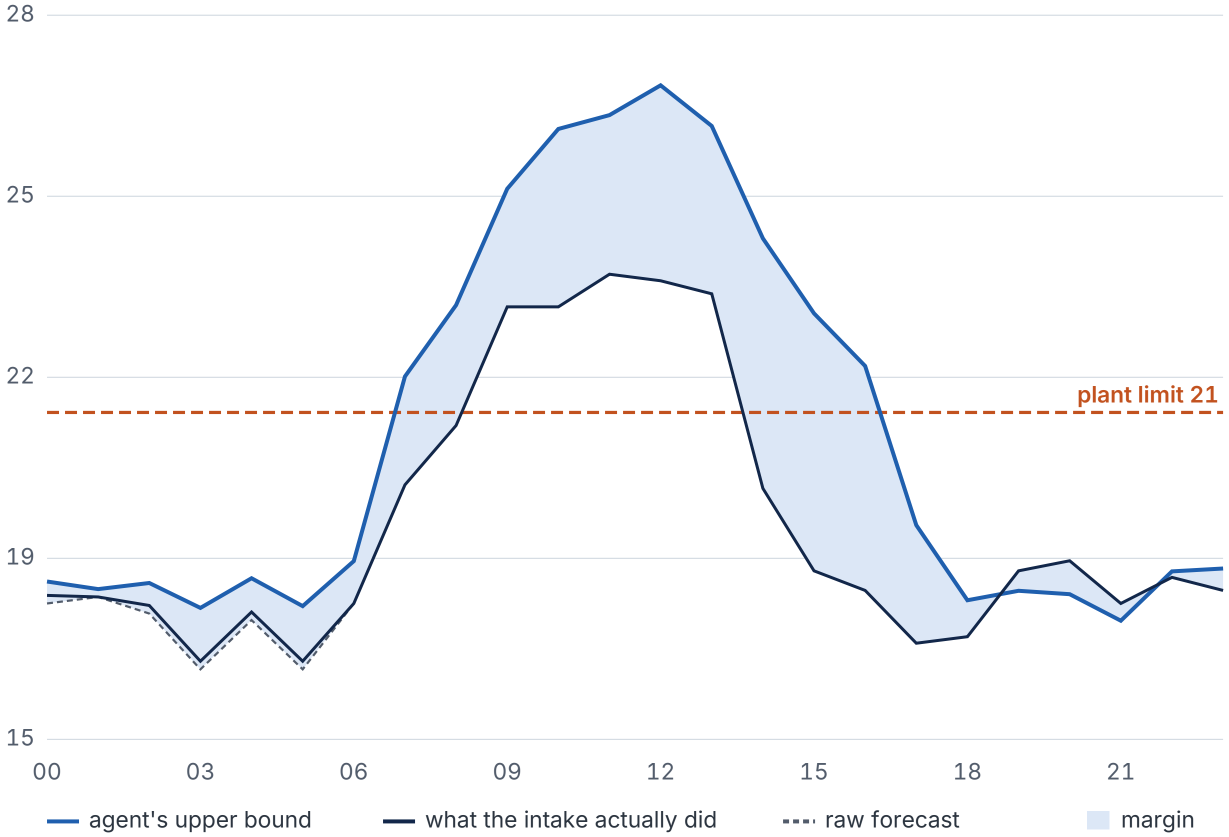
ESRI World Imagery. 0.3-0.5 m resolution shows objects, not nameplates. Cannot certify unit type or measure heights. Outlines are OpenStreetMap ways 1280344157 and 1386017492 on an independently georeferenced basemap, so an edge can sit a few metres off the roofline; every distance in this report is computed from the OpenStreetMap geometry, not from the picture.

The findings

Charts first. The prose under each one says only what the chart cannot.

The bound against the plant limit

degrees Celsius at the intake



The pale band is the safety margin: the distance the agent adds on top of the raw forecast before it will commit to an hour. The blue line is the resulting upper bound, the navy line is what the intake temperature actually did, and the dashed line is the plant's limit.

Read the crossings: the actual went above the bound in 3 of the 24 hours, which is what page 6 measures directly. All 3 were hours the agent had already put on chillers, and **0 of the hours it released for free cooling exceeded the plant limit**. A bound exceeded on an hour the agent did not act on costs nothing; the hours that matter are the ones it said yes to.

Chiller runtime, agent against incumbent

hours of mechanical cooling across 913 days the agent never trained on

14.3% less chiller runtime, a share so it holds at any hall size

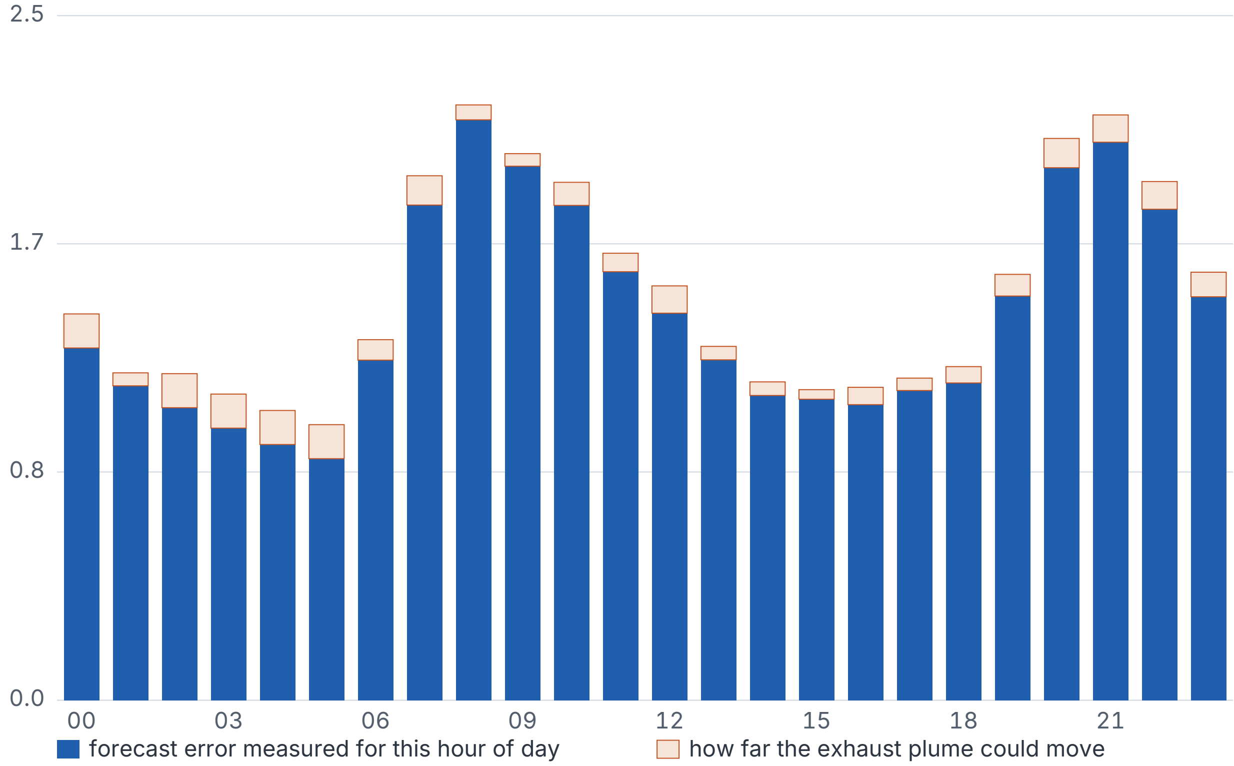
The comparison is against a tuned reactive controller using an on-site sensor, which is what operators verifiably run today, not against doing nothing. **Held-out** means days the agent never trained on: the record was split chronologically and the second half, 896 days, was kept back.

How the decision is made

Each hour the agent takes the forecast for this building's own tile, adds a margin, and compares the result against the plant's limit. It commits to free cooling only when the **upper bound** clears the **limit**, not when the forecast does.

Where each degree of the margin comes from

degrees Celsius added on top of the forecast



The margin is measured, not chosen. The lower part of each bar is the forecast error this agent has actually made at that hour of the day (**group-conditional**: the error is measured per hour of day rather than pooled across all of them, because a 3 a.m. forecast and a 3 p.m. forecast are not equally hard). The upper part is how far the exhaust plume could move if the wind sits differently from the direction planned for.

The gates an hour has to pass

Gate	Setting here	What it protects
Plant limit	21.0 °C	the intake temperature the hall is committed to
Dew point	15.0 °C	condensation on cold surfaces inside the hall
Notice	3 h	the plant cannot change mode instantly
Switch budget	2 per day	chillers and dampers wear out when cycled
Minimum dwell	3 h	a mode has to hold before another change is allowed

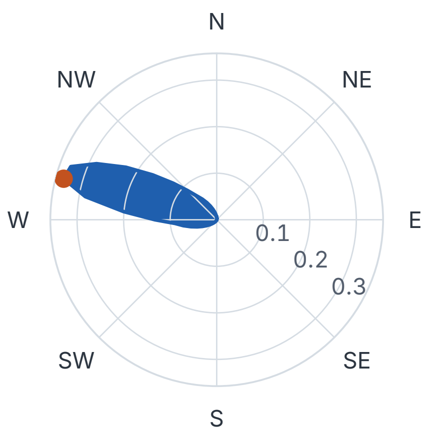
The switch budget is the reason a cold hour can still run chillers. With 2 changes permitted in a day, spending one early costs the agent the ability to switch later, so it schedules the whole day at once by dynamic programming rather than deciding each hour as it arrives.

What this document is · the interface recomputes for any configuration a reader selects. This file was generated for the one named above, chosen as the configuration where this site's agent does the most while still changing mode at least once. If the screen disagrees with this page, the configuration differs.

The physics

A data centre's condensers blow hot air out of the building. When the wind sits wrong that exhaust drifts toward a neighbour's air intake, so the machine can end up breathing warmed air while the forecast still reports the ambient. The agent therefore does not trust the forecast at the intake until it has solved where the exhaust actually goes.

Intake rise by wind bearing



worst 0.34 °C at 285°, from 576 solves

The solve	
bearings swept	72, every 5°
wind speeds	8, 0.5 to 12 m/s
solves	576
solver	GPU (NVIDIA Warp)
wall clock	7.46 s
bearings refused	0 of 72
worst rise	0.34 °C at 285°
mean rise	0.028 °C
condenser bank	100 m of facade, 20 cells

576 solves, on this building's own outline. Every one of the 72 wind bearings is solved across a grid of wind speeds, on the footprints taken from OpenStreetMap rather than an idealised box. The worst case anywhere on the compass is **0.34 °C** of intake rise, at 285 degrees.

That is a small number until the bound sits on the limit, and then it decides the hour. Including this term moved unsafe hours from **3 to 3** across the five-year record and **added** free-cooling hours at the same time, because knowing where the exhaust goes lets the agent say yes on the bearings that carry it away.

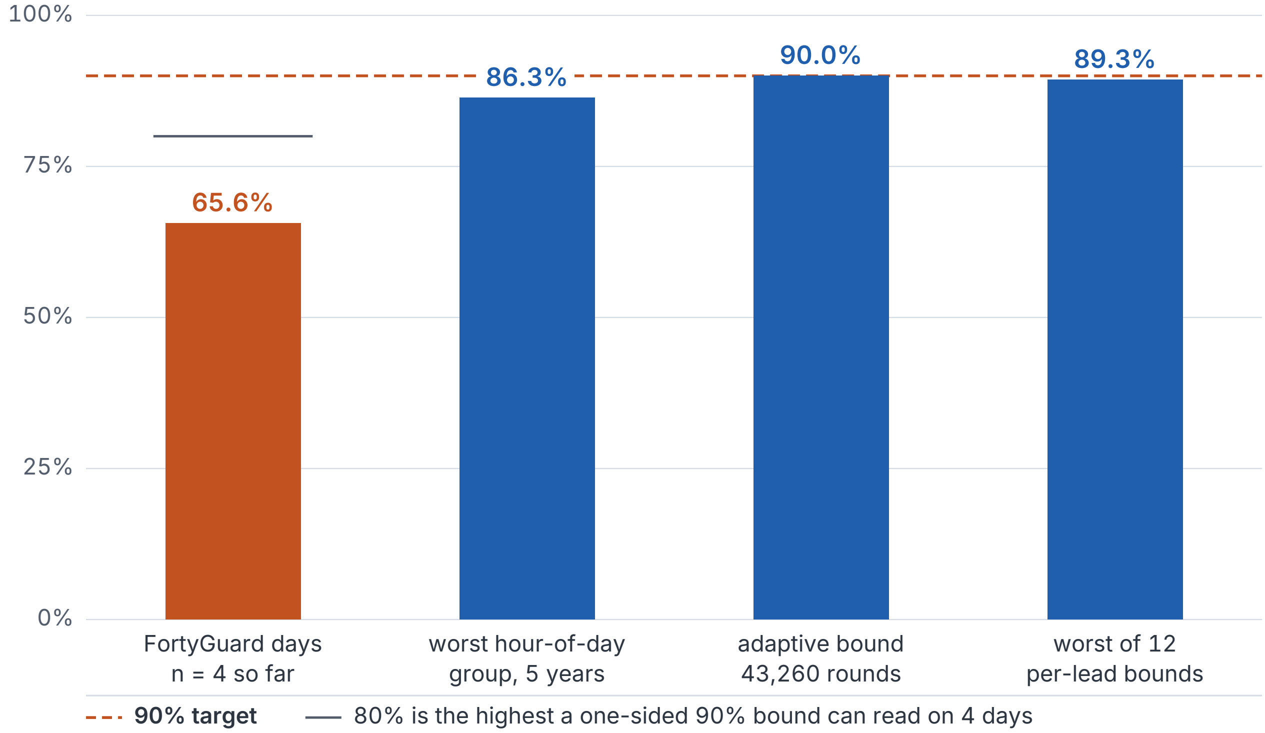
Why this needs a GPU: a single solve is not the workload, the bound is. Stating "90% of the time the intake stays under X" means running the physics many times over a spread of conditions. **A hundred of these solves take about a minute on a processor and under a second on the GPU through NVIDIA Warp, and the two agree to within a ten-thousandth of a degree.** This site's table took **7.5 seconds** on GPU (NVIDIA Warp).

At this placement no bearing is refused: nothing blocks the path between the exhaust and the intake, so all 72 directions return a number the agent is willing to use. The refusal is a property of geometry, not of this site, and where it does fire it is expensive: elsewhere in the sweep it costs the agent thousands of hours a year, and that cost is published rather than hidden.

Validation, and how the bound is measured

Bound coverage, by population size

share of hours the real intake stayed under the bound



A **bound** is a number the agent commits to the real intake staying under. Coverage is the share of hours it did. The target is 90%, and each population below is as large as the record currently allows.

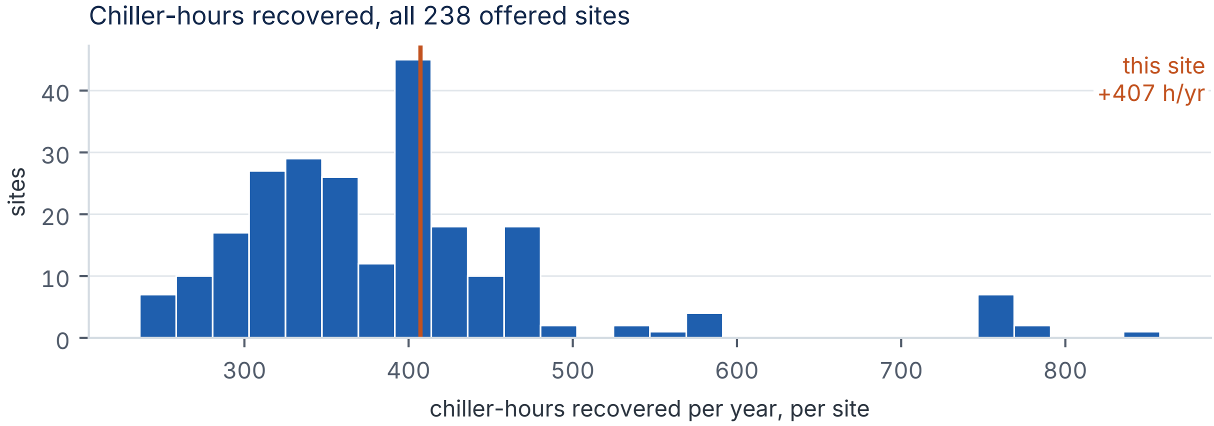
What the measurement shows

- The method reaches its target **wherever the record is large enough to measure it**. Twelve separately calibrated per-lead bounds, every one at or above 90%, the worst at 89%. An adaptive bound over 42,227 rounds of the five-year record realised 90%.
- **The FortyGuard-specific calibration currently rests on 4 measured forecast days**, and reads 66%. A one-sided 90% bound needs **9** such days: at 4 the arithmetic ceiling is 80%, so the target is a function of how many days have been forecast rather than of the method.
- A calibration day is **earned, not bought**. It is a forecast plus the day that follows it, so the count rises by one for every further day the site is forecast on. Nine days of operation is the whole distance between the figure above and the target.
- Conditioning already recovers **most of the gap**. Measuring the error per hour of day rather than pooling every hour together lifts the worst group from 74% to 86%.

What was tested	Volume	Result
Real recorded weather	42,730 h / 1,792 d	the record the schedule was scored on
Held-out days	896 d	never trained on
Re-plans compared	21,359	90% changed nothing at all
Free-cooling hours taken	16,559	15 unsafe, 0.91 per thousand

Scale, and what it is worth

One hall is a pilot. The agent covers **238 sites**, each built on its own OpenStreetMap footprint, its own station weather record and its own tariff, drawn from 639 tagged data centre facilities mapped across the United States.



This is every site the agent is offered on. **12 more were measured and excluded** because the agent's own constraints made it worse than the incumbent there, and those are not sold without site-specific engineering work on them first: at that geometry the safety margin hands back more free-cooling hours than it wins, so the agent runs the chillers more than the controller it would replace, the worst of them by 3,649 hours a year. They are not in the chart, because an axis wide enough to hold them would compress everything above into a sixth of the frame; they are counted here instead, and they stay on the map marked measured rather than ready. A portfolio that reports where it does not work is one a reader can believe about where it does.

Who buys this

An operator running many data centres, and the engineer inside it who owns the energy number. Their tenants' service level agreements commit them to a maximum supply air temperature, so an excursion is a contract breach and not just a warm hour, which is why the conservative buffer exists in the first place. What replaces the buffer has to come with a bound and a published breach rate, or it does not get switched on.

At this site	Value	Basis
Measured footprint	78,638 m²	OpenStreetMap rings, the same outlines the solver runs on
IT load	55 to 110 MW	DERIVED from a watts per square metre density
Recovered hours	+407 h/yr	backtested
Value of those hours	$305,846 to $1,254,023 /yr	MODELED. Measured hours times published rates times derived megawatts
The rate behind it	$5,543 to $11,365 /MW-IT/yr	32 cells: published tariffs x published chiller efficiencies, swept

The percentage is the honest headline and the dollars are the illustration. 14% less chiller runtime needs no assumption about how big the building is; the money rows do, and the two that depend on one are labelled.

Appendix: every hour, and each reason once

The 24 hours of this day carry **4 distinct reasons** between them. Each is explained once below, with the hours it covers and the tightest of those hours as the worked example.

Blocked by air temperature 10 hours: 07:00 to 16:00

The air was simply too warm. In each of these hours the agent's upper bound on intake temperature sat above the plant's 21.0 °C limit, so free cooling would have risked the hall. It is the bound that is compared against the limit, never the raw forecast: the margin is what makes the comparison safe rather than optimistic.

Worked example, 07:00 · ambient air **19.78 °C**, less the **0.3 °C** the forecast earns over assuming today repeats, plus 0.22 °C of plume rise at the forecast wind bearing, plus a measured **1.91 °C** margin, gives a **21.61 °C** bound against the 21.0 °C limit. The wind then sat somewhere slightly different, so the intake itself reached 19.78 °C: covering that difference is what the plume half of the margin is for.

Free cooling ran 8 hours: 00:00 to 06:00, 17:00

Free cooling ran. In each of these hours the agent's upper bound cleared the plant limit with the full measured margin already added, so outside air did the cooling and the chillers stayed off.

Worked example, 17:00 · ambient air **17.11 °C**, plus **0.82 °C** where the forecast read warmer than assuming today repeats, plus a measured **1.17 °C** margin, gives a **19.1 °C** bound against the 21.0 °C limit. With no neighbouring hall to warm the incoming air, the intake is the ambient air, and it reached 17.11 °C.

Blocked by the dew-point gate 4 hours: 18:00 to 21:00

Temperature passed and humidity did not. Pulling in air this damp risks condensation on cold surfaces inside the hall, so the dew-point gate blocks the hour independently of how cool the air is.

Worked example, 19:00 · ambient air **18.33 °C**, less the **1.89 °C** the forecast earns over assuming today repeats, plus a measured **1.55 °C** margin, gives a **17.99 °C** bound against the 21.0 °C limit. With no neighbouring hall to warm the incoming air, the intake is the ambient air, and it reached 18.33 °C.

Blocked by the switch budget 2 hours: 22:00 to 23:00

These hours were **cold enough to free-cool and were run on chillers anyway**, and the reason is the schedule rather than the weather. The plant permits 2 mode changes a day. Spending one here would leave the agent unable to switch when a longer or colder block arrives, so the whole day is planned at once and these hours are the ones the plan gives up. This is the explanation a thermostat cannot give, because a thermostat has no plan to be constrained by.

Worked example, 23:00 · ambient air **18 °C**, less the **1.19 °C** the forecast earns over assuming today repeats, plus a measured **1.56 °C** margin, gives a **18.37 °C** bound against the 21.0 °C limit. With no neighbouring hall to warm the incoming air, the intake is the ambient air, and it reached 18 °C.

The full schedule

hh	mode	safe	reason	bound	limit	actual	margin
00	free	yes	-	18.2	21.0	17.9	1.4
01	free	yes	-	18.0	21.0	17.9	1.2
02	free	yes	-	18.1	21.0	17.7	1.2
03	free	yes	-	17.7	21.0	16.8	1.1
04	free	yes	-	18.2	21.0	17.6	1.1

hh	mode	safe	reason	bound	limit	actual	margin
05	free	yes	-	17.7	21.0	16.8	1.0
06	free	yes	-	18.5	21.0	17.8	1.3
07	mechanical	no	dry-bulb	21.6	21.0	19.8	1.9
08	mechanical	no	dry-bulb	22.8	21.0	20.8	2.2
09	mechanical	no	dry-bulb	24.8	21.0	22.8	2.0
10	mechanical	no	dry-bulb	25.8	21.0	22.8	1.9
11	mechanical	no	dry-bulb	26.0	21.0	23.3	1.6
12	mechanical	no	dry-bulb	26.5	21.0	23.2	1.5
13	mechanical	no	dry-bulb	25.8	21.0	23.0	1.3
14	mechanical	no	dry-bulb	23.9	21.0	19.7	1.2
15	mechanical	no	dry-bulb	22.7	21.0	18.3	1.1
16	mechanical	no	dry-bulb	21.8	21.0	18.0	1.1
17	mechanical	yes	-	19.1	21.0	17.1	1.2
18	mechanical	no	dew point	17.8	21.0	17.2	1.2
19	mechanical	no	dew point	18.0	21.0	18.3	1.5
20	mechanical	no	dew point	17.9	21.0	18.5	2.0
21	mechanical	no	dew point	17.5	21.0	17.8	2.1
22	mechanical	yes	switch budget	18.3	21.0	18.2	1.9
23	mechanical	yes	switch budget	18.4	21.0	18.0	1.6

What would change the answer

The hours above that ran chillers on a named schedule constraint were not held by the weather. Re-planning this same day with the switch budget raised from **2 to 4** releases **2 of the 24 hours** to free cooling. Re-planning it with the minimum dwell cut from 3 h to 1 h releases **0**. On this day and this configuration the binding constraint is the number of mode changes the plant permits, not its dwell requirement and not the air.

Change to the plant's own limits	Hours released	Which hours
switch budget 2 per day to 4	+2	22:00 to 23:00
minimum dwell 3 h to 1 h	+0	none

That is a re-plan, not a promise. The switch budget exists because cycling chillers and dampers wears them out, so the hours above are available only if the plant is willing to pay that wear, and the agent deliberately does not make that trade on the operator's behalf. What it does is price it: this is the number an engineer needs in order to decide whether the setting is worth revisiting. Each of the 2 is re-planned and re-checked in `explain.py` before it may be printed here.

Generated by `AGENTIC-ARBITER/src/site_report.py` from this site's own artefacts. Every figure here is read from the file that produced it; the rows labelled MODELED are the only ones resting on an unmeasured assumption.